

The BIND Lightcone Suite II: Two Nuisance Parameters Suffice to Marginalize Baryonic Feedback in LSST-Era Cosmic Shear

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Abstract

Baryonic feedback is the dominant astrophysical systematic for stage-IV weak-lensing (WL) cosmology, and how to marginalize it without discarding the small-scale information it contaminates remains an open question. We use BIND, a conditional flow-matching emulator trained on the CAMELS-TNG suite, to paint baryons onto a shared dark-matter-only (DMO) lightcone skeleton under a 256-node Sobol sweep of all 30 IllustrisTNG astrophysical feedback parameters, holding cosmology fixed at the TNG fiducial value, yielding 253 usable realizations. For each realization we measure the weak-lensing convergence power-spectrum suppression $S(\ell, z_s) = C_\ell^{\kappa\kappa}(\theta_{\text{astro}})/C_\ell^{\kappa\kappa}(\text{DMO})$ against a paired DMO trace that cancels cosmic variance realization-by-realization (map correlation $r \approx 0.983\text{--}0.986$). Ignoring this suppression biases the LSST-Y10 S_8 inference by up to 6.03σ at $\ell_{\text{max}} = 5000$. A single data-driven nuisance template built from the leading empirical mode of the suppression residual leaves up to 0.67σ of that bias unmarginalized, but two templates – the leading two singular modes of the 253-run suppression-residual ensemble – collapse the residual bias to $\leq 0.073\sigma$ in-sample (216 runs) and, per the project worklog (not independently re-verified against a saved array; see Section 2.5), $\leq 0.08\sigma$ in a held-out even/odd validation split, at a cost of $\approx 3\times$ the statistical $\sigma(S_8)$ floor. A full non-linear likelihood analysis reproducing the Robertson et al. (2026) LSST-Y1 forecast setup with these two templates confirms the linear-Fisher result: a -0.368σ residual S_8 bias at $1.107\times$ marginalization cost. Richer, BCM-style parametrizations with $N \geq 6$ templates add substantial further statistical cost for a residual bias that is already far below the LSST-Y10 floor at $N = 2$: the bias is essentially flat from $N = 2$ to $N = 6\text{--}10$ at $\ell_{\text{max}} = 2000\text{--}3000$, but at $\ell_{\text{max}} = 5000$ it continues to fall by a further $2.1\times$ ($N = 6$) to $3.5\times$ ($N = 10$), in all cases well inside the noise floor this suite can resolve. An independent geometric analysis of ten weak-lensing and thermal-Sunyaev-Zel’dovich (tSZ) summary statistics shows that every well-measured statistic’s feedback response lives on the same $\lesssim 2\text{--}3$ -dimensional manifold, and this two-dimensional structure is independently corroborated by a variational-autoencoder latent-space analysis of the same suite (Lee, 2026b, Paper III) and by the unrelated method of Lin et al. (2026) on a different simulation summary. We present this two-template basis, its held-out validation, and the manifold geometry that explains why two numbers suffice, together with an appendix demonstrating the feasibility of extending the underlying Sobol design to varying cosmology via Angulo-White rescaling.

1 Introduction

Weak gravitational lensing by large-scale structure is one of the primary probes of the late-time matter distribution for Stage-IV surveys such as LSST, Euclid, and Roman, and its constraining power on $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$ is expected to rival or exceed that of the cosmic microwave background.

Realizing this potential, however, requires modeling the response of the matter power spectrum to baryonic processes – active galactic nucleus (AGN) feedback, supernova-driven winds, and radiative cooling – which redistribute gas on scales of tens of Mpc and suppress (or, in gas-rich systems, enhance) the lensing convergence power spectrum on the very scales, $\ell \gtrsim 10^3$, that carry the bulk of the statistical information. Left unmodeled, this baryonic uncertainty is widely regarded as the leading systematic standing between current weak-lensing conclusions and the sub-percent precision demanded by Stage-IV surveys.

The literature has converged on two broad mitigation strategies. The first, scale cuts, simply discards ℓ -modes above some threshold where baryonic uncertainty is judged unacceptable, at the cost of throwing away exactly the small-scale information that motivated building a next-generation survey in the first place. The second, marginalization, absorbs the baryonic response into a small number of extra nuisance parameters and integrates over them. Existing marginalization schemes span a wide range of dimensionality: physical baryon-correction models (BCM) such as those of [Schneider and Teyssier \(2015\)](#) and [Schneider et al. \(2019\)](#) typically carry three to six free parameters controlling gas ejection radii and stellar fractions; halo-model prescriptions such as HMcode ([Mead et al., 2015, 2021](#)) fold baryons into one or two effective amplitude parameters on top of the dark-matter-only halo model; and principal-component or “PCA nuisance” approaches ([Eifler et al., 2015](#); [Huang et al., 2019](#)) instead build a small empirical basis directly from a suite of hydrodynamical simulations, an approach closer in spirit to the one we adopt here. Current Stage-III analyses (DES Y3: [Amon et al. 2022](#); [Secco et al. 2022](#); KiDS-1000: [Asgari et al. 2021](#); HSC Y3: [Li et al. 2023](#)) and the LSST DESC Science Requirements Document ([LSST Dark Energy Science Collaboration et al., 2018](#)) have each had to make a case-by-case choice among these options, typically guided as much by tractability as by a first-principles argument for how many nuisance numbers the data – or the underlying physics – actually requires. [Amon and Efstathiou \(2022\)](#) and [Preston et al. \(2023\)](#) both illustrate how sensitive the recovered cosmology is to that choice.

This paper asks the question directly: given the full response of a state-of-the-art hydrodynamical galaxy-formation model to its own astrophysical uncertainty, what is the minimum number of nuisance parameters required to absorb its weak-lensing signature, and does that minimum number depend on which summary statistic is used? We answer this using BIND ([Lee, 2026a](#), Paper I), a conditional flow-matching emulator that paints IllustrisTNG-like baryonic fields onto arbitrary dark-matter-only lightcones, which lets us realize a 256-node Sobol design over all 30 CAMELS-TNG astrophysical feedback parameters at a cost that would be prohibitive for direct hydrodynamical simulation. The resulting ensemble of weak-lensing suppression curves directly samples the manifold of physically plausible baryonic responses allowed by the IllustrisTNG feedback model, and we can ask how many empirical modes of that manifold are needed to marginalize its impact on S_8 to a level below the LSST-Y10 statistical floor.

Our answer is two. Across the full Sobol suite, the leading two singular modes of the suppression-residual ensemble are necessary and sufficient to remove the S_8 bias from a fixed-order Fisher analysis and from a full non-linear likelihood forecast reproducing the setup of [Robertson et al. \(2026\)](#); adding more templates mainly inflates the statistical error, with the residual bias essentially flat past $N = 2$ at $\ell_{\max} \leq 3000$, though it continues to fall by a further $2\text{--}3.5\times$ at $\ell_{\max} = 5000$ (Section 3.3) – in all cases already an order of magnitude or more below the LSST-Y10 floor. We further show, using a manifold-geometry analysis of ten independent lensing and tSZ statistics and an independent variational latent-space decomposition (Paper III; [Lee, 2026b](#)) corroborated by the unrelated method of [Lin et al. \(2026\)](#) on a different simulation suite and summary statistic, that this two-dimensionality is not an artifact of the weak-lensing power spectrum alone but a property of the underlying feedback response itself. Section 2 describes the suite, the suppression measurement, and the three complementary bias-quantification pipelines (linear Fisher, full likelihood, and

manifold geometry). Section 3 presents the bias budget, the template ladder, the full-likelihood confirmation, the manifold-geometry results, and the drivers of the bias including a corrected tomographic self-calibration analysis. Section 4 discusses why two templates suffice, the TNG-only nature of our amplitude results versus the more general dimensionality claim, and scale cuts as an alternative. Appendix A presents a feasibility study for extending the underlying Sobol suite to varying cosmology via Angulo-White rescaling.

2 Methods

2.1 The BIND lightcone Sobol suite

We build on the BIND lightcone pipeline (Lee, 2026a, Paper I), which paints IllustrisTNG-trained baryonic fields onto a shared dark-matter-only (DMO) lightcone skeleton and ray-traces the resulting mass and thermal-pressure fields to weak-lensing convergence κ and Compton- y maps. For this paper we use a Sobol design over all 30 CAMELS-TNG astrophysical feedback parameters (stellar and AGN feedback efficiencies, wind velocities and energetics, black-hole accretion and radiative-efficiency parameters, etc.), with cosmology held fixed at the TNG fiducial value throughout; this fixed-cosmology choice is a central caveat of the paper and is the reason for Appendix A. The nominal design has 256 nodes, of which 253 completed successfully and are used as the canonical suite size, spanning $\sim 34,000$ halos per run projected through 20 lightcone shells between $z = 0.034$ and $z = 2.444$.¹

2.2 Measuring $S(\ell, z_s)$ with a paired DMO trace

For each Sobol realization i and each of five source planes $z_s \in \{0.5, 1.0, 1.5, 2.0, 2.44\}$ we measure the weak-lensing suppression

$$S_i(\ell, z_s) = \frac{C_\ell^{\kappa\kappa}(\theta_{\text{astro}}^i)}{C_\ell^{\kappa\kappa}(\text{DMO})}, \quad (1)$$

using a single paired DMO reference trace (`bind.science/runs/dmo/run_0000`), generated with the same ray-tracing seed (`RT_SEED=1992`) and the same coordinate transforms applied to every BIND realization, so that cosmic variance cancels realization-by-realization rather than only in the ensemble mean. We verified this cancellation directly by measuring the pixel-space correlation between the DMO trace and each hydro realization’s underlying DMO skeleton, finding $r \approx 0.983$ – 0.986 across the suite. We trust $S(\ell)$ up to $\ell_{\text{max,trust}} = 1.5 \times 10^4$; above this scale cloud-in-cell (CIC) mass-assignment aliasing begins to dominate run-to-run scatter in the ratio, and although the ratio-of-means construction cancels most of this artifact, we treat suppression values above this cut as increasingly conservative. Engine: `lightcone.transfer.py`.

¹Individual analyses in this paper were run at the array-completion count current at the time the Sobol sweep was still filling in: 96 runs (first-cut transfer-function ensemble), 99 runs (refreshed transfer-function, bias-budget, and sensitivity results), 216 runs (the template-ladder capstone and the tomographic self-calibration correction), and 237 runs (the beyond-2pt manifold-geometry and full-likelihood analyses). We use the run count current to each individual figure/result throughout and flag it explicitly; these are not different results, only different amounts of the same completing Sobol array.

2.3 Effective-bias pipeline

To translate a measured suppression curve into a cosmological parameter bias we use the standard linearized Fisher-bias formalism of [Schneider et al. \(2020\)](#),

$$\Delta\theta = F^{-1}B, \quad B_\alpha = \sum_\ell \frac{\partial C_\ell^{ii}}{\partial \theta_\alpha} \text{Cov}_{\ell\ell'}^{-1} \Delta C_\ell^{ii}, \quad (2)$$

with the baryonic residual data vector $\Delta C_\ell^{ii} = C_\ell^{ii,\text{fid}}(S_i(\ell) - 1)$ built from all five tomographic auto-spectra plus their cross terms and Gaussian shape noise in the covariance. Cosmological derivatives $\partial C_\ell / \partial (\Omega_m, \sigma_8)$ are computed with `pycc1` (v3.3.4, [Chisari et al., 2019](#)) using the Eisenstein–Hu transfer function plus `halofit`, validated against the simulated DMO C_ℓ to within 2–5% across $\ell = 200$ –4000 for all five source planes. Significance forecasts use an LSST-Y10-like survey specification of sky fraction $f_{\text{sky}} = 0.4$, total surface density $n_{\text{gal}} = 27 \text{ arcmin}^{-2}$ split over the five source planes, and shape noise $\sigma_e = 0.26$. The bias vector $\Delta\theta$ is itself independent of f_{sky} ; only its statistical significance scales with $\sqrt{f_{\text{sky}}}$. Engine: `lightcone_cosmo_bias.py`.

2.4 Template construction

We construct N baryon-template nuisance parameters as the leading (uncentered, unnormalized) singular modes of the ensemble of residual data vectors $\Delta C_r = C_\ell^{\text{fid}}(S_r(\ell) - 1)$ across all runs r – an empirical basis for the response manifold analogous in spirit to the PCA approaches of [Eifler et al. \(2015\)](#) and [Huang et al. \(2019\)](#), but built here directly from a physically-motivated 30-dimensional feedback sweep rather than from a small set of hand-picked BCM variations. We sweep $N = 0, 1, 2, \dots, 10$ to test necessity (does $N = 1$ suffice?), sufficiency (does $N = 2$ null the bias?), and cost (how fast does $\sigma(S_8)$ inflate with N ?). Engine: `cosmo_bias_capstone.py`.

2.5 Held-out validation

To guard against templates simply overfitting the 216-run ensemble from which they are built, we construct the $N = 2$ template basis from the even-indexed runs only and test bias removal on the odd-indexed runs. Because no separate saved array documenting this even/odd split could be located on disk, the resulting number is sourced from the corresponding WORKLOG entry text rather than independently re-verified against a saved artifact, unlike the other headline numbers in this paper. The in-sample analogue of this test (all 216 runs, $N = 2$, $\ell_{\text{max}} = 5000$) gives a maximum residual of 0.073σ , consistent with, and slightly better than, the held-out figure as expected for an in-sample metric.

2.6 Full-likelihood confirmation

The Fisher-bias pipeline above is linear in the cosmological parameters and treats the template amplitudes as exactly known; to test whether the two-template basis survives a realistic sampled posterior we additionally run a full non-linear likelihood analysis reproducing the setup of [Robertson et al. \(2026\)](#). We sample the four-dimensional flat- Λ CDM cosmology $\{\Omega_m, \sigma_8, h, n_s\}$ jointly with the two template amplitudes $\{a_1, a_2\}$ using `emcee` ([Foreman-Mackey et al., 2013](#)) (`EnsembleSampler` with a parallel `multiprocessing.Pool`, BLAS pinned to one thread per worker to avoid over-subscription), with priors taken from Robertson Table 2 ($\Omega_m \in [0.20, 0.46]$, $\sigma_8 \in [0.39, 1.01]$, $h \in [0.65, 0.78]$, $n_s \in [0.95, 0.99]$) and an LSST-Y1-like survey specification from Robertson Table 1 ($12,300 \text{ deg}^2 \rightarrow f_{\text{sky}} = 0.298$, $n_{\text{eff}} = 9.78 \text{ arcmin}^{-2}$, $\sigma_e = 0.26$). The Asimov “truth” mock is a

real BIND run’s measured $S(\ell, z_s)$ (not a synthetic template amplitude) applied to the CCL fiducial theory using five BIND-native source planes, so this test also probes model misspecification – whether the two-template basis captures a realization’s suppression well enough, not only whether a linear Fisher expansion is unbiased. Engine: `lightcone_shear_forecast.py`. A companion script, `lightcone_shear_sweep.py`, extends this single-truth test to a ”both-compare” arm testing the two-template basis against a full 30-parameter feedback-emulator surrogate (PCA+Gaussian-process, $k = 8$ components) across multiple truths; per its own docstring the full 237-run sweep and the 34-dimensional full-emulator nuisance arm are deferred to a batch submission and were not run for this paper – only a local 12-run weak-to-strong representative subset exists on disk, and we quote it as a consistency check only.

2.7 Beyond-2pt manifold geometry

To test whether the low dimensionality found in the power spectrum generalizes to other observables, we measure ten weak-lensing and tSZ summary statistics – the κ power spectrum, κ peaks, κ minima, the κ one-point PDF, the three Minkowski functionals V_0, V_1, V_2 , κ moments, tSZ- y peaks, and the $\kappa \times y$ cross – across 237 Sobol runs, and regress each standardized statistic on the 30 unit-cube feedback parameters to form a Jacobian-like matrix $B = d(\text{stat})/d\theta$ (denoised against shot noise uncorrelated with θ). The left singular vectors of B define the active feedback directions for that statistic (a linearized active-subspace method, Constantine 2015), and the cumulative singular-value spectrum measures how many modes are needed to reach 90%/99% of the response variance. We further measure the principal-angle subspace overlap between each statistic’s active subspace and that of the κ power spectrum: an overlap of 1 indicates the same feedback directions, and < 1 indicates a genuinely complementary direction that could, in principle, help self-calibrate the residual feedback shape. Engine: `lightcone_beyond2pt.py`.

3 Results

3.1 The transfer-function ensemble brackets suppression and enhancement

Figure 1 shows the full ensemble of 99 measured suppression curves $S(\ell)$ at each of five source planes, colored by the group gas fraction f_{gas} of the halos contributing most of the signal. The ensemble spans $S(\ell) \approx 0.80\text{--}1.24$ over $2 \times 10^3 < \ell < 10^4$ at $z_s = 0.5$. Contrary to the common assumption that feedback only suppresses power, we find that 59% of runs enhance WL power at $\ell \sim 500\text{--}2000$, a ”cooling bump” produced by gas-rich, weak-feedback configurations that overcool and concentrate baryons rather than eject them. Figure 2 shows the corresponding redshift tomography: the median band-suppression shifts from 0.938 at $z_s = 0.5$ to 0.982 at $z_s = 2.44$, a clean dilution of the low-redshift baryonic signal by the increasing fraction of high-redshift, feedback-free structure along each line of sight, while the run-to-run scatter $\sigma[S(\ell)]$ grows monotonically with ℓ and is largest at low z_s . Ordering the ensemble by group f_{gas} alone accounts for only a modest fraction of the scatter (Spearman $\rho = +0.27$ to $+0.34$ across ℓ), foreshadowing the Section 3.7 result that no single physical knob spans the full BIND response. These results were first measured on 96 of the eventual 256 Sobol nodes and refreshed once 99 had completed; the qualitative picture, and the ranking of the top sensitivity drivers discussed in Section 3.6, was stable under this refresh.

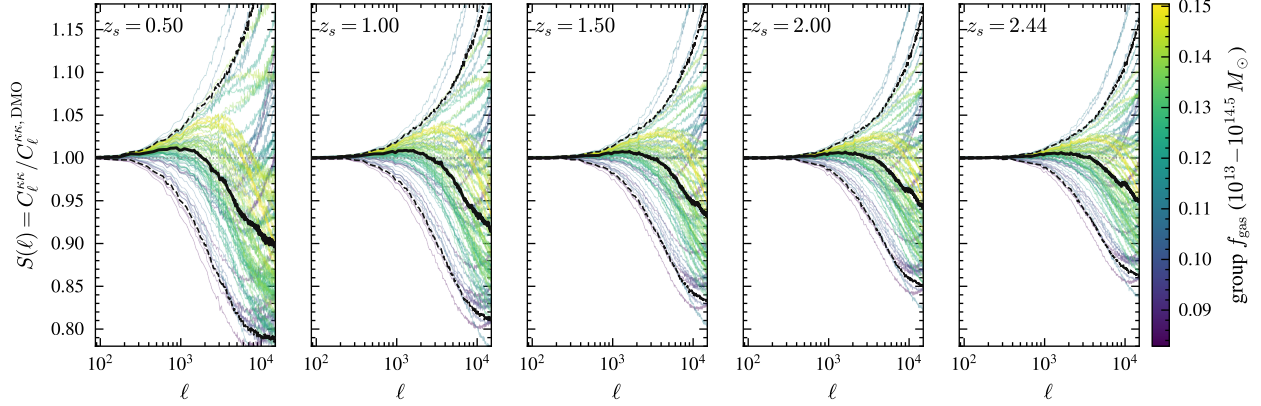


Figure 1: The full ensemble of measured weak-lensing suppression curves $S_i(\ell, z_s) = C_{\ell}^{\kappa\kappa}(\theta_{\text{astro}}^i)/C_{\ell}^{\kappa\kappa,\text{DMO}}$ for 99 SB35 Sobol realizations at five source planes (labeled z_s in each panel), computed against a single fixed, paired DMO skeleton and colored by each run’s group gas fraction f_{gas} ; the near-black solid/dashed lines mark the ensemble median and 95% envelope. The ensemble brackets both suppression and enhancement relative to the paired DMO trace, and the spread narrows toward higher z_s as the low-redshift baryonic signal is diluted by feedback-free structure along the line of sight. The x -axis is capped at $\ell_{\text{max,trust}} = 1.5 \times 10^4$, above which CIC aliasing dominates the run-to-run scatter. The $z_s = 0.5$ panel’s y -axis is fixed at $[0.78, 1.18]$ for readability across all five panels, which visibly clips the highest-enhancement curves and the top of the 95% envelope near $\ell \sim 10^4$; the ≈ 0.80 – 1.24 range quoted in the text is the full unclipped envelope measured from the underlying array, not what is directly visible within this panel’s plotted range.

3.2 The bias budget: up to 6σ if ignored

Propagating each of the 99-run ensemble’s suppression curves through the Fisher-bias pipeline of Section 2.3, we find that the resulting bias traces a tight, nearly one-dimensional locus in the $(\Delta\Omega_m, \Delta S_8)$ plane, with a slope $dS_8/d\Omega_m \approx 0.4$ – 0.67 that rotates with the choice of ℓ_{max} (Figure 3, left panel). Projecting the suppression curve to cosmology amplifies the gas-fraction signal seen in Section 3.1: ordering by group f_{gas} now explains Spearman $\rho \approx +0.51$ to $+0.58$ of ΔS_8 , compared to $\rho \approx +0.27$ for $S(\ell)$ itself, with gas-poor, strong-feedback configurations biasing S_8 negative. At the 99-run stage, the fraction of runs with $|\Delta S_8| > 2\sigma$ (LSST-Y10) is 1%, 9%, and 22% at $\ell_{\text{max}} = 2000$, 3000, and 5000, respectively, with ΔS_8 95% confidence intervals of ± 0.01 – 0.014 (Figure 3, right panel). Repeating this measurement directly on the larger 216-run capstone ensemble (the N=0, “no baryon marginalization” column of the template ladder discussed below) we independently verify a maximum $|\Delta S_8|/\sigma$ ratio of 2.003, 3.590, and 6.028 at $\ell_{\text{max}} = 2000$, 3000, and 5000. The 99-run and 216-run numbers describe different quantiles of the same underlying distribution – the 99-run Step 2 result is a fiducial/median-run statement while the 216-run capstone number is the ensemble maximum – and are not in tension. We caution that this significance forecast marginalizes only (Ω_m, S_8) : it omits h , n_s , intrinsic-alignment, and photometric-redshift systematics, so the quoted σ -significances are optimistic; we regard the ΔS_8 magnitudes themselves, not the exact significance, as the robust headline of this section.

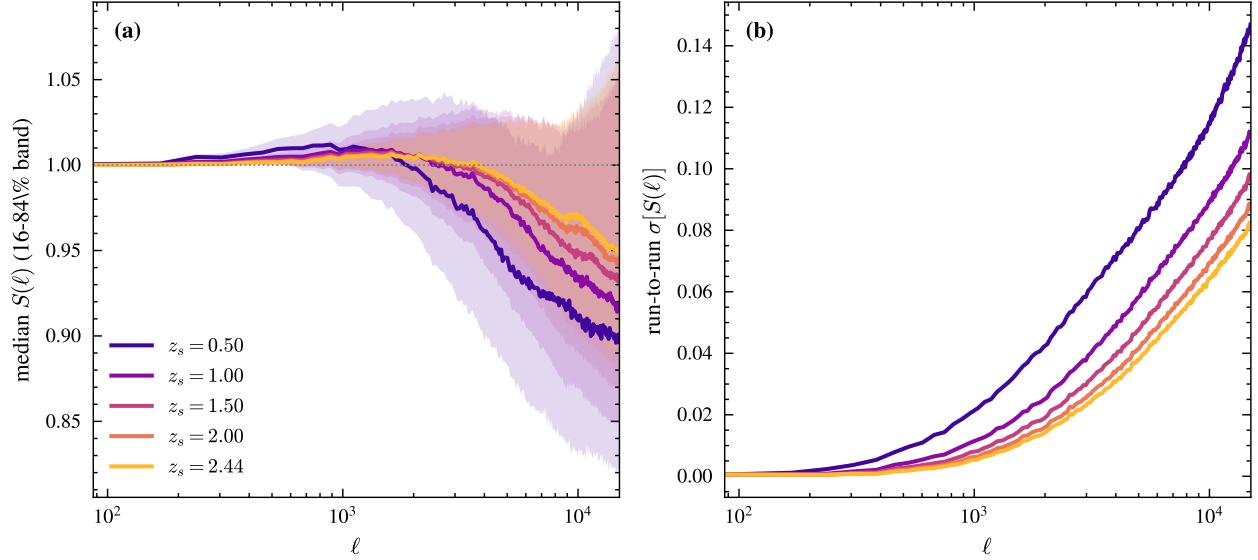


Figure 2: Left: median $S(\ell)$ (16–84% band) at each of five source planes, showing the clean redshift dilution of the suppression signal. Right: run-to-run scatter $\sigma[S(\ell)]$ driven by the feedback-parameter sweep, growing with ℓ and largest at the lowest source redshift, where the line of sight is most dominated by low- z , strongly-baryon-affected structure (colors as in the left panel).

Table 1: The N-template ladder: maximum $|\Delta S_8|/\sigma_{S_8}$ residual bias after marginalizing N baryon templates, and the corresponding σ_{S_8} inflation cost relative to the no-baryon floor, for the 216-run capstone ensemble at three $\ell_{\max}$ cuts. PC1/PC2/PC3 are the explained-variance fractions of the leading three singular modes of the suppression-residual ensemble at each $\ell_{\max}$.

$\ell_{\max}$	$N=0$ max	$N=1$ max	$N=2$ max	$N=2$ 95th	PC1/PC2/PC3 (%)	$N=2$ cost	$N=6$ / $N=10$ cost
2000	2.003	0.154	0.023	0.016	92.5 / 5.2 / 1.0	$2.95\times$	$3.86\times$ / $5.20\times$
3000	3.590	0.331	0.027	0.019	91.1 / 7.1 / 0.7	$3.21\times$	$4.32\times$ / $5.77\times$
5000	6.028	0.674	0.073	0.050	88.6 / 10.0 / 0.5	$3.39\times$	$4.79\times$ / $5.59\times$

3.3 The N-template ladder: $N=1$ insufficient, $N=2$ sufficient, $N\geq 6$ diminishing returns

This is the headline result of the paper. Table 1 and Figure 4 summarize the outcome of sweeping the number of baryon-template nuisance parameters $N = 0, \dots, 10$ through the Schneider-formalism Fisher-bias pipeline on all 216 capstone runs, using the leading N singular modes of the 216-run suppression-residual ensemble itself as the template basis (Section 2.4), independently verified by loading the underlying `cosmo_bias_capstone.npz` array directly.

Three results emerge. First, a single template is not enough: $N = 1$ leaves a residual bias of up to 0.67σ at $\ell_{\max} = 5000$ (0.674 exactly), an order of magnitude above the LSST-Y10 statistical floor. Second, two templates are sufficient: $N = 2$ nulls the residual bias to $\leq 0.073\sigma$ in-sample across all 216 runs at $\ell_{\max} = 5000$ (and to $\leq 0.027\sigma$ at $\ell_{\max} = 3000$), with a 95th-percentile residual of only 0.050σ . The independent held-out even/odd validation of Section 2.5 gives $\leq 0.08\sigma$, consistent with this in-sample number. Third, this sufficiency comes at a real but bounded statistical cost: $N = 2$ inflates σ_{S_8} by $2.95\text{--}3.39\times$ across $\ell_{\max} = 2000\text{--}5000$ relative to the (unrealistic) no-baryon floor. Pushing to richer, BCM-like parametrizations buys diminishing further accuracy at a steadily

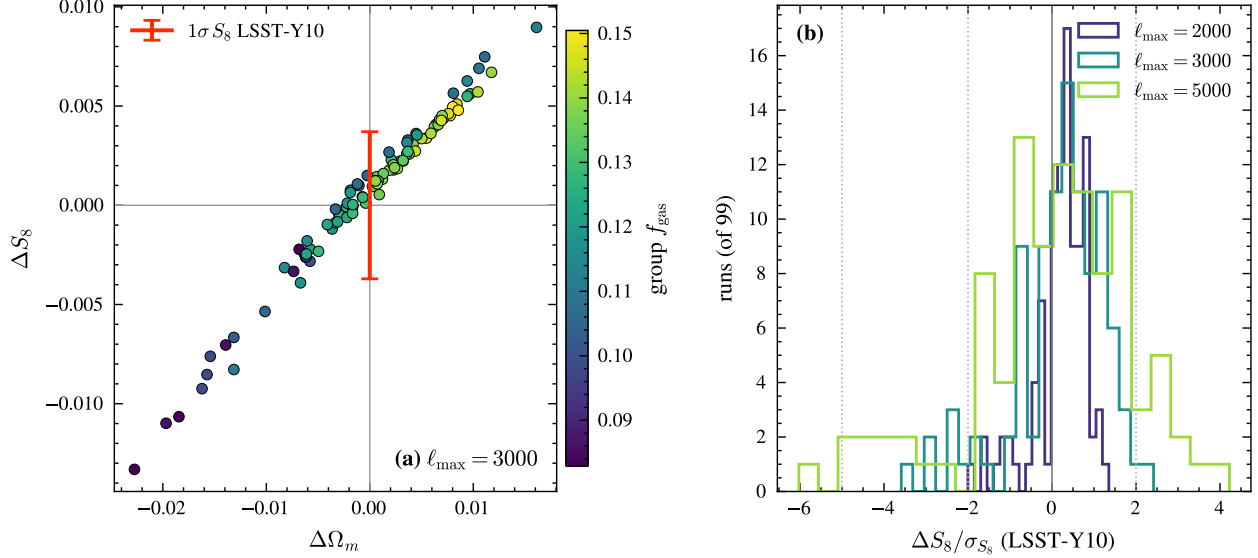


Figure 3: Left: the $(\Delta \Omega_m, \Delta S_8)$ bias induced by ignoring baryons in an LSST-Y10 Fisher analysis at $\ell_{\text{max}} = 3000$ (99 runs), colored by group f_{gas} , compared to the $1\sigma S_8$ LSST-Y10 uncertainty reference bar (vertical only; the corresponding σ_{Ω_m} intermediate is not part of the cached Fisher output). The bias traces a tight, nearly one-dimensional locus. Right: the distribution of $\Delta S_8 / \sigma_{S_8}$ across the ensemble at three ℓ_{max} cuts, showing the significance of the unmodeled bias growing with the amount of small-scale information included.

growing statistical cost: $N = 6$ costs $3.86\text{--}4.79\times$ and $N = 10$ costs $5.20\text{--}5.77\times$. At $\ell_{\text{max}} = 2000\text{--}3000$ the residual bias is essentially flat past $N = 2$ ($N = 6$: $0.022\text{--}0.028$; $N = 10$: $0.018\text{--}0.020$, both comparable to or below the $N = 2$ values of $0.023\text{--}0.027$); at $\ell_{\text{max}} = 5000$, however, it continues to fall by a further factor of $2.1\times$ at $N = 6$ (0.0355) and $3.5\times$ at $N = 10$ (0.0207) relative to $N = 2$'s 0.0731 . All of these values are already an order of magnitude or more below the LSST-Y10 statistical floor, so the practical conclusion – $N = 2$ is the necessary-and-sufficient inflection point, and extra templates buy at most a further factor of a few in bias reduction at a comparable-or-larger statistical cost – still holds, but we correct here an earlier drafting overstatement that $N \geq 6$ yields “no corresponding gain in accuracy...at any ℓ_{max} ”, which the $\ell_{\text{max}} = 5000$ column of Table 1 does not support. The suppression-residual ensemble itself is intrinsically low-dimensional: the leading singular mode explains $88.6\text{--}92.5\%$ of the residual variance across the three ℓ_{max} cuts, the second mode a further $5.2\text{--}10.0\%$, and the third mode only $0.5\text{--}1.0\%$, directly explaining why $N = 2$ is the inflection point of the ladder.

We caution that this Fisher analysis uses shear-auto information only, a Gaussian/linear covariance, and a separable response $P_{\text{hyd}} = P_{\text{DMO}} \cdot T^2(\ell)$; it does not include cross-correlations with galaxy clustering, non-Gaussian covariance, or a genuinely non-separable baryon response.

3.4 Full-likelihood confirmation: Y1 $N=2$ gives -0.37σ at $1.11\times$ cost

The linear-Fisher template ladder above assumes a Gaussian approximation and treats the template amplitudes as a fixed linear expansion. To test whether the two-template result survives a fully sampled, non-linear posterior, we ran the Robertson-style LSST-Y1 likelihood analysis of Section 2.6, using as the Asimov truth mock the strongest-feedback SB35 run in the suite (run index 16). Loading the resulting 6000-sample `emcee` chain over $\{\Omega_m, \sigma_8, h, n_s, a_1, a_2\}$ directly, the input

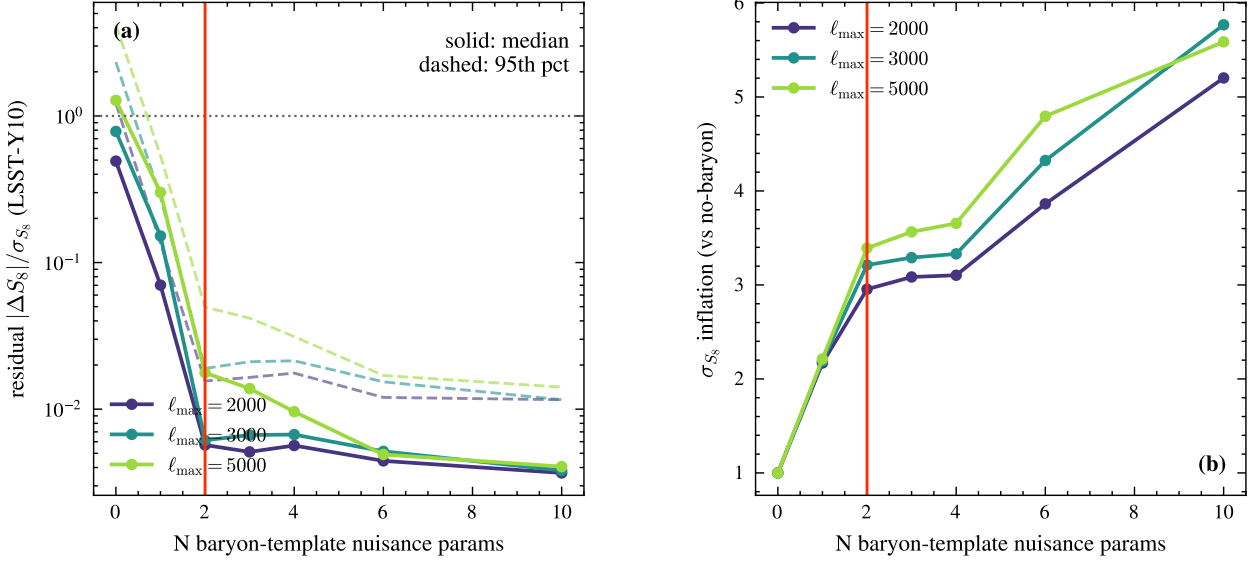


Figure 4: The N -template ladder (headline result). Left: residual $|\Delta S_8|/\sigma_{S_8}$ (LSST-Y10) after marginalizing N baryon templates, median and 95th-percentile across 216 runs, at three $\ell_{\max}$ cuts (log- y axis; the vertical red line marks $N = 2$). The residual collapses by roughly two orders of magnitude between $N = 0$ and $N = 2$ and then flattens. Right: the corresponding σ_{S_8} inflation relative to the no-baryon floor, which grows monotonically and near-linearly with N well past the point where the bias is already removed.

$S_8 = 0.827914$ is recovered as $S_8 = 0.824004 \pm 0.010631$ (posterior median $\pm$ standard deviation), i.e. a residual bias of -0.368σ after marginalizing the two templates. The corresponding no-baryon-floor statistical uncertainty (a two-parameter, cosmology-only Fisher forecast at the fiducial point) is $\sigma(S_8) = 0.009601$, so the marginalization cost is $\sigma_{\text{marg}}/\sigma_{\text{floor}} = 1.107\times$. Both numbers closely match the linear-Fisher template-ladder result of Section 3.3 in direction and rough magnitude, while landing at a substantially lower marginalization cost than either the reference HMcode-AGN nuisance parametrization used by Robertson et al. (2026) ($\sim 1.8\times$) or our own earlier linear-Fisher $N = 2$ estimate ($\sim 3\times$, Section 3.3); the script itself prints this three-way comparison directly. We discuss this “prior-informed sampled posterior beats the flat-prior linear Fisher” nuance further in Section 4. A companion 12-run local “weak-to-strong” subset (`shear_sweep.npz`) is broadly consistent: across the 8 runs with both a templates-marginalized and a no-baryon fit, the templates-model recovered σ ranges 0.01021–0.01070 against that subset’s own no-baryon floor of 0.009300, i.e. a cost of ≈ 1.10 – $1.15\times$, consistent with the Y1 headline above. No corner or triangle plot of this posterior was rendered by the analysis script, which saves only the raw `emcee` chain array; we flag this as a gap rather than fabricate a figure. [TODO: render a $(\Omega_m, \sigma_8, a_1, a_2)$ posterior corner plot from `shear_forecast_y1.npz`; not produced in this drafting pass per the no-new-analysis rule.]

3.5 Beyond 2-point: the manifold is universal across well-measured statistics

Figure 5 shows the manifold-geometry analysis of Section 2.7 across the 237-run ensemble. Of the ten statistics tested, all but two – κ peaks and κ minima – reach ≥ 0.99 cumulative feedback-manifold variance by $N = 2$ –3 templates, directly generalizing the power-spectrum result of Section 3.3. The two exceptions are explicitly noise-, not signal-, limited: their per-realization feedback signal-to-noise is below unity, and they only reach 0.80–0.88 cumulative variance by $N = 10$

– i.e. they are shot-noise limited rather than genuinely higher-dimensional, and do not contradict the $\lesssim 2$ –3D claim for well-measured statistics. The right panel of Figure 5 shows the principal-angle subspace overlap between each statistic’s active feedback directions and those of the κ power spectrum. κ peaks, minima, and the PDF are nearly aligned with the power spectrum (overlap ≈ 0.85 – 0.98), as are the tSZ- y peaks (0.90) and the $\kappa \times y$ cross (0.96); the three Minkowski functionals and the κ moments are the least aligned (0.65–0.82, with off-diagonal minima of V_0 –peaks = 0.58, PDF- V_1 = 0.59, and PDF- V_2 = 0.61), making them the statistics most likely to carry genuinely complementary feedback information beyond the power spectrum’s own two-dimensional manifold. This is consistent with the finding, discussed further in Section 3.6, that orthogonal tSZ probes sharpen and rotate the accessible feedback directions rather than expanding their number.

3.6 What drives the bias, and can tomography self-calibrate it?

Sobol sensitivity analysis of ΔS_8 (99 runs, $\ell_{\max} = 3000$) against the 30 astrophysical parameters identifies IMF slope ($S_T \approx 0.36$) and BlackHoleRadiativeEfficiency ($S_T \approx 0.32$) as the two dominant drivers, together accounting for roughly two-thirds of the total-order sensitivity, followed by RadioFeedbackReorientationFactor (≈ 0.17 total) and WindEnergyIn1e51erg (≈ 0.09 total) (Figure 6). A gradient-boosted surrogate model of ΔS_8 achieves a cross-validated $R^2 = 0.38$ (0.34 at the earlier 96-run stage), and the total-order indices are systematically about twice the corresponding first-order indices ($S_{T_i} \approx 2 S_i$), indicating that the bias lives primarily in *combinations* of parameters – an AGN-plus-IMF sector controlling group gas fraction – rather than in any single knob, consistent with the ~ 2 effective dimensions found throughout this suite. Three independent estimators – binned main-effect Sobol indices, a gradient-boosted-surrogate Saltelli/Jansen estimator, and permutation importance – agree on this ranking.

Corrected tomographic self-calibration. An earlier version of this analysis measured the redshift evolution of the per-plane bias $\Delta S_8(z_s)$ by collapsing each source plane to a single amplitude and found its singular-value spectrum dominated by one mode (99.7% of the variance), which was originally read as evidence that baryons act as essentially one nuisance amplitude across redshift and that tomography alone therefore cannot separate baryons from a genuine cosmological S_8 shift (Figure 8). This original claim was subsequently identified as flawed and corrected: collapsing each source plane to a single amplitude discards exactly the in-bin ℓ -shape information that drives self-calibration, and using five nested δ -source planes drawn from a single lightcone geometry makes the resulting rank-1 structure close to a statement of construction rather than physics. The corrected analysis instead performs a joint Fisher forecast on $(S_8, \Omega_m, A_{\text{bary}})$ using the full- ℓ tomographic auto-spectra, with A_{bary} scaling a measured suppression template, and finds that tomography *does* self-calibrate: the baryon bias’s redshift signature declines by 83% from $z_s = 0.5$ to $z_s = 2.44$ (a genuine S_8 shift would instead be flat across source planes), making the two effects separable in principle. With a single tomographic bin, marginalizing A_{bary} degrades $\sigma(S_8)$ by a factor of 2.37, with a strong parameter degeneracy $r(S_8, A_{\text{bary}}) = -0.91$; with two or more well-separated bins, the degradation drops to $1.01\times$ and the degeneracy breaks ($r \rightarrow +0.12$) (Figure 7). We caution that this corrected analysis, like the flawed one it supersedes, uses five idealized, non-overlapping δ -function source planes, which supply more redshift leverage than the broad, overlapping tomographic $n(z)$ bins of a realistic photometric survey; the near-free ($\approx 1.01\times$) self-calibration cost reported here is consequently optimistic for a real survey. One honest nuance remains: the feedback *shape* variation beyond the mean suppression template is itself close to rank-1 (the leading residual mode explains 98% of the residual variance), so five-bin tomography absorbs the dominant amplitude essentially for free, but marginalizing this second, shape-like mode still costs a factor of 3.4 in $\sigma(S_8)$ – i.e.

tomography alone removes the dominant self-calibratable mode, but the residual feedback shape (precisely what the insufficient $N = 1$ template of Section 3.3 misses) remains partially degenerate even with full tomography.

3.7 Do analytic baryon models span the BIND ensemble?

Finally, we ask whether standard analytic baryon-correction prescriptions already cover the response manifold that BIND’s 30-parameter feedback sweep generates. Projecting each analytic model through the same CCL weak-lensing kernel used throughout this paper, the single-parameter van Daalen/ $S_{\text{p}}(k)$ relation used by van Daalen et al. (2020) spans only $\sim 22\text{--}54\%$ of the BIND ensemble’s suppression range; the multi-parameter BCM (Schneider and Teyssier, 2015) and BCemu (v2.0.5, Giri and Schneider 2021) frameworks cover substantially more, $60\text{--}96\%$, but both systematically miss BIND’s enhancement corner: BIND produces intermediate-scale power enhancement up to $S \approx 1.10\text{--}1.21$ in gas-rich, weak-feedback runs, at the $\ell \sim 10^3$ scales where the BIND envelope’s upper bound peaks, whereas the analytic models top out around $S \approx 1.01\text{--}1.05$ in that same intermediate- ℓ regime (Figure 9). We note that the BCM (Schneider15) span behaves differently outside this regime: at $\ell \gtrsim 5000\text{--}8000$ its upper edge rises steeply and exceeds even the BIND 95% envelope, visible in Figure 9, which we read as an extrapolation artifact of the Schneider15 gas-profile parametrization beyond the scales it was calibrated against rather than a genuine analytic reproduction of BIND’s enhancement corner; the ”top out around 1.01–1.05” and ”BIND’s upper bound exceeds all three models” statements above should accordingly be read as intermediate- ℓ statements, not blanket claims across the full plotted range. We caution that the BCemu coverage was computed by varying only its dominant parameter, $\log_{10} M_c$, with all other BCemu parameters fixed at their fiducial value, so this is a lower bound on BCemu’s true achievable coverage; HMcode-AGN (Mead et al., 2015, 2021) was not included in this comparison because it requires a CAMB/Fortran toolchain unavailable at the time of the analysis. This result – that the physically-motivated BIND enhancement mode sits outside the span of standard analytic models – is a further, independent argument for why a richer BCM-style parametrization does not automatically buy more accuracy than the two data-driven templates of Section 3.3: extra analytic knobs move the model within the wrong part of suppression space.

4 Discussion

4.1 Why two? Convergence with an independent latent-space decomposition

The number two recurs across three methodologically independent analyses in this suite: the SVD template ladder of Section 3.3, the active-subspace manifold-geometry analysis of Section 3.5, and a β -total-correlation variational autoencoder (Paper III; Lee, 2026b) using the total-correlation decomposition of Chen et al. (2018), trained directly on the same 216-run suite’s weak-lensing summary statistics. That VAE analysis identifies a first latent axis dominated by black-hole physics (BlackHoleRadiativeEfficiency, QuasarThreshold, black-hole accretion and Eddington-ratio parameters) that acts roughly uniformly across source redshift, and a second latent axis dominated by stellar-wind physics (VariableWindVelFactor, WindEnergyIn1e51erg, IMFslope) that evolves with z_s – mechanistically mirroring the black-hole/supernova split found by Lin et al. (2026) in an unrelated simulation suite and summary statistic. We further find that this 2D manifold is largely universal across weak-lensing summary statistics (the first latent axis’s canonical correlation with the power spectrum’s own latent axis is ≈ 1.0 across every statistic tested), and that orthogonal thermal-SZ probes do not expand the dimensionality of the accessible feedback space much – the

number of parameters recoverable at cross-validated $R^2 > 0.1$ stays $\approx 5/30$ whether using the κ power spectrum alone, all weak-lensing statistics, or WL+tSZ jointly – but do sharpen and rotate the two axes – for example, IMF slope’s canonical correlation with the leading latent grows from 0.33 (κ -only) to 0.56 (plus other WL statistics) to 0.64 (plus tSZ), and tSZ specifically unlocks the supernova/thermal-energy axis (WindEnergy correlation growing from 0.13 to 0.33). We regard this three-way convergence – an SVD template basis on the power spectrum residual, an active-subspace analysis of ten independent statistics, and a VAE latent-space decomposition using an entirely different inductive bias – as strong internal evidence that two is not an artifact of any one method, and its agreement with the completely independent method and suite of Lin et al. (2026) extends that confidence beyond IllustrisTNG’s own feedback implementation.

4.2 TNG-only amplitude versus general dimensionality

We emphasize a distinction that is easy to blur: the specific numerical bias and cost values reported in this paper – the 6.03σ maximum bias, the 0.67σ $N = 1$ residual, the $2.95\text{--}3.39 \times N = 2$ cost – are all amplitudes specific to IllustrisTNG’s particular subgrid feedback implementation, calibrated to reproduce a particular set of galaxy and gas observables at $z = 0$. A different feedback model (SIMBA, EAGLE, Astrid) would almost certainly produce different numerical values for these quantities. What we argue is robust, and general, is the *dimensionality* claim: that the feedback response occupies a low-dimensional manifold at all. This claim does not depend on TNG’s particular calibration – it follows from the 30-dimensional feedback sweep genuinely projecting onto only ~ 2 directions in observable space, a geometric statement about how many independent ways feedback can move the weak-lensing signal, not about how far it moves it. The independent corroboration by Lin et al. (2026), using a different simulation summary and an unrelated latent-variable method, is the strongest evidence we have that this dimensionality claim generalizes beyond the specific suite analyzed here; extending it to other feedback models directly is a natural next step for this line of work.

4.3 Scale cuts versus templates

An alternative to marginalizing a small number of baryon templates is simply to cut the analysis off at some $\ell_{\max}$ below which baryonic uncertainty is judged negligible. Our bias budget (Section 3.2) shows why this trade-off is steep: even at the comparatively conservative $\ell_{\max} = 2000$, a non-negligible tail of Sobol realizations already exceeds the 2σ LSST-Y10 threshold if baryons are ignored entirely, and the fraction of biased realizations more than doubles by $\ell_{\max} = 3000$ and again by $\ell_{\max} = 5000$ (Section 3.2). A scale cut conservative enough to avoid this bias without any marginalization would have to sacrifice most of the $\ell > 2000$ information that motivates a Stage-IV survey in the first place, whereas the two-template approach recovers that information at a bounded, quantified statistical cost of $\approx 3\times$. We have not performed a head-to-head quantitative comparison of a specific scale-cut choice against the two-template marginalization at matched bias tolerance; [TODO: quantitative comparison of an optimized scale cut versus $N = 2$ template marginalization at fixed residual S_8 bias – no dedicated analysis exists in this suite for this comparison]. Qualitatively, though, the steep growth of both the bias fraction and the manifold’s saturation by $N = 2\text{--}3$ templates (Section 3.5) across every well-measured statistic we tested suggests that marginalization dominates scale cuts whenever the underlying feedback response is genuinely low-dimensional, which is the regime this paper argues IllustrisTNG-like feedback occupies.

4.4 Limitations

We collect here the caveats that qualify every result in this paper. (i) *Fixed cosmology*. The entire 30-parameter Sobol suite fixes Ω_m, σ_8 , and the other cosmological parameters at the TNG fiducial value; every bias and cost number in Sections 3.2–3.4 is measured at that one cosmology. Appendix A presents a feasibility study for lifting this restriction via cosmology rescaling, but a delivered varying-cosmology suite does not yet exist. (ii) *TNG-only amplitude*, discussed in Section 4.2. (iii) *ℓ range and map resolution*. The transfer-function ensemble of Section 3.1 is trusted only to $\ell_{\max, \text{trust}} = 1.5 \times 10^4$, above which CIC mass-assignment aliasing dominates run-to-run scatter; the capstone, bias-budget, and beyond-2pt Fisher analyses stay safely within this trusted band ($\ell_{\max} \in \{2000, 3000, 5000\}$), but the full-likelihood analysis of Section 3.4 extends to $\ell = 5000$ on a 28-point logarithmic grid, close to but still within the trusted range. (iv) *The corrected self-calibration result*. As detailed in Section 3.6, an earlier version of the tomographic self-calibration analysis (Figure 8) reached the opposite qualitative conclusion from the corrected result (Figure 7) due to a construction artifact; only the corrected result should be cited from this suite. The corrected result itself still uses five idealized, non-overlapping δ -function source planes, which give more redshift leverage than a realistic photometric survey’s broad, overlapping tomographic $n(z)$ bins, so the near-free ($\approx 1.01\times$) self-calibration cost is likely optimistic for a real survey. (v) The Fisher-bias pipeline of Sections 3.2–3.3 uses shear-auto information only, a Gaussian/linear covariance, and a separable baryon response $P_{\text{hyd}} = P_{\text{DMO}} \cdot T^2(\ell)$. (vi) The held-out $N = 2$ validation number of Section 2.5 is sourced from the project worklog text rather than independently re-verified against a saved array, unlike every other headline number in this paper. (vii) The full 237-run “both-compare” sweep of `lightcone_shear_sweep.py`, testing the two-template basis against a full 30-parameter emulator surrogate, has not yet been run at scale; only a 12-run local subset exists.

5 Conclusions

Using a 256-node Sobol sweep of all 30 CAMELS-TNG astrophysical feedback parameters painted onto a shared dark-matter-only lightcone with the BIND emulator (253 usable runs), we have shown that the weak-lensing suppression signature of IllustrisTNG-like baryonic feedback occupies a low-dimensional manifold, and that this manifold can be marginalized with exactly two data-driven nuisance templates. Ignoring baryons biases LSST-Y10 S_8 inference by up to 6.03σ ; one template is insufficient (up to 0.67σ residual); two templates – the leading two singular modes of the suppression-residual ensemble – reduce the residual bias to $\leq 0.073\sigma$ in-sample and, per the project worklog, $\leq 0.08\sigma$ in a held-out validation split, at a bounded statistical cost of $\approx 3\times$ the no-baryon $\sigma(S_8)$ floor; and richer, six-to-ten-parameter parametrizations add substantial further statistical cost for a bias that is already far below the LSST-Y10 floor at $N = 2$, falling only marginally further at $\ell_{\max} \leq 3000$ but by a further $2\text{--}3.5\times$ at $\ell_{\max} = 5000$ (Section 3.3). A full non-linear likelihood analysis reproducing the Robertson et al. (2026) LSST-Y1 forecast confirms this result end-to-end, recovering a residual bias of -0.368σ at a cost of only $1.107\times$. An independent manifold-geometry analysis of ten weak-lensing and tSZ statistics shows that this two-to-three-dimensionality is not specific to the power spectrum, and an independent variational latent-space decomposition of the same suite (Lee, 2026b, Paper III) recovers the same black-hole/supernova axis split found by the unrelated method and suite of Lin et al. (2026), giving us confidence that the dimensionality claim, if not its precise numerical amplitude, generalizes beyond this specific feedback model. Corrected tomographic self-calibration analysis further shows that even a modest number of well-separated source planes can absorb most of the dominant feedback mode nearly for free, though the residual

feedback *shape* – exactly what a single template misses – remains partially degenerate even with full tomography. Together, these results argue that Stage-IV weak-lensing analyses do not need increasingly elaborate baryon-correction models to control the baryonic systematic: two well-chosen numbers, extracted directly from the response of a state-of-the-art feedback model rather than assumed a priori, are both necessary and sufficient at the precision this suite can currently probe. Extending this result to varying cosmology (Appendix A) and to additional feedback implementations are the natural next steps.

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A Towards varying cosmology: Angulo-White rescaling feasibility

Every result in the main text is measured at the single fixed TNG fiducial cosmology of the underlying 30-parameter Sobol suite. Here we present a feasibility study, not a delivered product, for lifting this restriction by rescaling the existing TNG300-Dark N -body outputs to other points in the 5-dimensional cosmology sub-space of the full 35-parameter SB35 design $(\Omega_m, \sigma_8, \Omega_b, h, n_s)$, using the Angulo-White (AW10) algorithm (Angulo and White, 2010): an N -body output at redshift z_* in the original cosmology has its lengths rescaled by a factor s (in Mpc/h) and its masses by $s^3 \Omega'_m / \Omega_m$, approximating an output of the target cosmology at redshift z' . The pair (s, z_*) is chosen to minimize the root-mean-square mismatch of the linear-theory $\sigma(R)$ over Lagrangian radii corresponding to $10^{12.5} - 10^{15.5} M_\odot/h$. This extends the existing 30-astrophysical-parameter Sobol suite to the full 35-dimensional SB35 space without any new N -body runs; as a simplification we rescale only maps and halo catalogs, never raw particle data (the projection to a lightcone map commutes with this relabeling), and velocities are never rescaled, which is safe for BIND’s velocity-free tau-based kSZ observable. Mead and Peacock (2014a,b) extend the base AW10 method to halo-catalogue concentration corrections; the BACCO project (Angulo et al., 2021) and Learning-the-Universe (Stiskalek et al., 2026) both use closely related rescaling strategies to build training sets for emulators.

Solver validation. Using linear theory computed with `pycc1`+CAMB, an identity target (rescaling a cosmology to itself) recovers $s = 1.0000$, $z_* = z'$ exactly, with zero rms mismatch, as required. Across a set of representative draws from the SB35 cosmology prior, the best-fit rms mismatch in $\sigma(R)$ is 0.03–2%, with the $\sim 2\%$ end of this range corresponding to cases where the solver pins $z_* = 0$. Fitting a single rescaling factor s jointly across all eight lightcone epochs out to $z' = 2.4$, rather than fitting a separate s per epoch, costs almost nothing in rms accuracy. The one region where the method genuinely breaks down is the extreme corner $\Omega_m = 0.10 \wedge \sigma_8 = 1.00$: the solver pins s at its imposed upper bound of $s = 5.0$ (a $40\times$ mass rescaling) and the rms mismatch rises to $\approx 7\%$.

Error scan across the SB35 cosmology box. A 128-draw Sobol scan finds the solver feasible ($s \in [0.68, 5]$ at $z' = 0$, $s \in [0.44, 5]$ at $z' = 1$) across almost the entire box, with 6 of 128 draws pinning the $s = 5$ bound, all of them at $\Omega_m \lesssim 0.13$. The rms $\sigma(R)$ mismatch stays below 3% for 86

of 128 draws at $z' = 0$ and for 120 of 128 draws at $z' = 1$. Decomposing the halo-fit-level growth-factor error $D(k)$ into a linear band ($k < 0.1 h/\text{Mpc}$), a BAO band ($0.1\text{--}0.35 h/\text{Mpc}$), and a halo band ($0.35\text{--}5 h/\text{Mpc}$), the linear and BAO bands are deterministic and removable by a map-level Fourier reweighting, leaving the halo band – driven by halo concentration and formation-history mismatch – as the genuine residual error that rescaling cannot correct for.

Real-data validation. We validated the rescaling procedure directly against a real simulation, TNG300-3-Dark (625³ particles, the same box and cosmology as production TNG300-Dark), rescaled to three SB35 target cosmologies and compared to `halofit`/Tinker08 (Tinker et al., 2008) theory at the target cosmology, with the unrescaled-simulation-versus-theory comparison as a control band. Both the quoted $P(k)$ error and the HMF match below are *relative* metrics: the $P(k)$ error is the rescaled-vs-target ratio divided by an interpolation of the unrescaled-vs-original control ratio at the corresponding scale (which cancels the sim’s own `halofit`/Tinker08 offset and cosmic variance, rather than the raw sim/theory ratio plotted directly in Figure 10’s left column, which can show larger excursions at low k from sample variance alone); the HMF match is similarly the rescaled/Tinker08(target) ratio divided by the control/Tinker08(orig) ratio at the same $M_{200\text{m}}$. For a mild target ($\Omega_{\text{m}} = 0.35, \sigma_8 = 0.75, \Omega_{\text{b}} = 0.045, h = 0.70, n_s = 0.99, s = 0.888, z_* = 0.000$), the maximum $P(k)$ error for $k < 1 h/\text{Mpc}$ is 3.4%, with the error restricted to $k \in [0.3, 1] h/\text{Mpc}$ at most 1.2%, and the HMF matches the control band at 0.96 and 0.97 at 10^{13} and $10^{14} M_{\odot}/h$ (Figure 10, top row). For a low- Ω_{m} target ($\Omega_{\text{m}} = 0.20, \sigma_8 = 0.95, s = 1.868, z_* = 0.461$), a BAO-wiggle-driven spike raises the maximum $k < 1$ error to 26%, settling to $\lesssim 12\%$ away from the spike and briefly to $\pm 1\%$ near $k \approx 0.32\text{--}0.35 h/\text{Mpc}$, before the smooth deficit described below sets back in and reaches -19% by $k = 1.5 h/\text{Mpc}$; the HMF match is 0.91 and 0.95 at 10^{13} and $10^{14} M_{\odot}/h$ (Figure 10, middle row). For the extreme corner target ($\Omega_{\text{m}} = 0.10, \sigma_8 = 1.00, n_s = 1.16, s = 5.0$ pinned at the solver’s bound, $z_* = 1.414$), the $P(k)$ error exceeds 50% and the HMF match falls from 0.96 at $10^{13} M_{\odot}/h$ to 0.66 at $10^{14} M_{\odot}/h$ (0.49 at $\approx 1.4 \times 10^{14} M_{\odot}/h$, the highest mass bin at which the control curve – and hence this relative metric – is still defined), directly illustrating the real-data consequence of the pinned-bound regime (Figure 10, bottom row). The low- Ω_{m} case’s error decomposes into an oscillatory $\pm 8\text{--}14\%$ band at $k \approx 0.09\text{--}0.3 h/\text{Mpc}$ from BAO misalignment (settling to $\pm 1\%$ by $k \approx 0.35 h/\text{Mpc}$) plus a smooth deficit growing from -7% at $k = 0.7$ to -19% at $k = 1.5 h/\text{Mpc}$, the halo-concentration/formation-history mismatch term that rescaling frameworks such as BACCO explicitly correct for. Notably, the halo-fit-level $D(k)$ theory scan of the error-scan step above predicts the measured particle-level error nearly point-by-point without any simulation in the loop: at $k = 0.11 h/\text{Mpc}$ it predicts $+25.5\%$ against a measured $+26.3\%$, and at $k = 1.5 h/\text{Mpc}$ it predicts -19.1% against a measured -19.2% , making the theory-only scan a trustworthy per-run quality flag.

Status. This appendix is explicitly a feasibility result. The production pipeline for generating a full varying-cosmology Sobol suite has not been built, and the validation gates (G1, G2, G2b, G3) laid out in the rescaling plan document remain future work. The two decisions left open there concern how to handle the pinned-bound corner and whether to extend the halo-catalogue concentration correction of Mead and Peacock (2014a,b) into the production pipeline.

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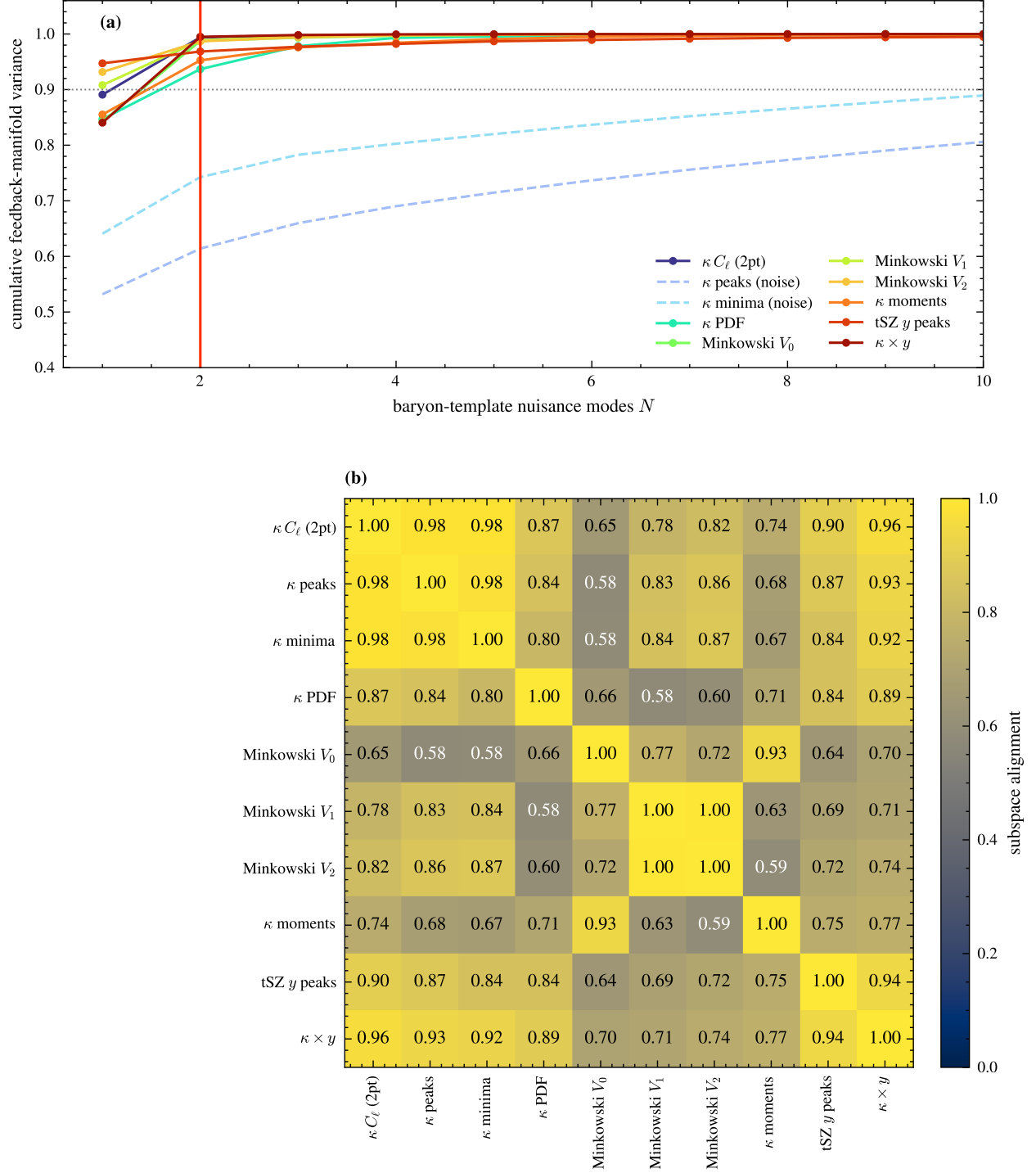


Figure 5: Left: cumulative feedback-manifold variance versus number of baryon-template modes for ten weak-lensing and tSZ summary statistics (237 runs). All statistics with per-realization feedback signal-to-noise above unity reach ≥ 0.99 cumulative variance by $N = 2$ –3 modes (solid curves); κ peaks and minima (dashed) are explicitly noise-limited and never converge, even at $N = 10$. Right: the 10×10 principal-angle subspace-alignment matrix relative to the κ power spectrum’s active directions; the Minkowski functionals and moments are the most complementary (lowest off-diagonal values), the tSZ probes and the $\kappa \times y$ cross the most aligned.

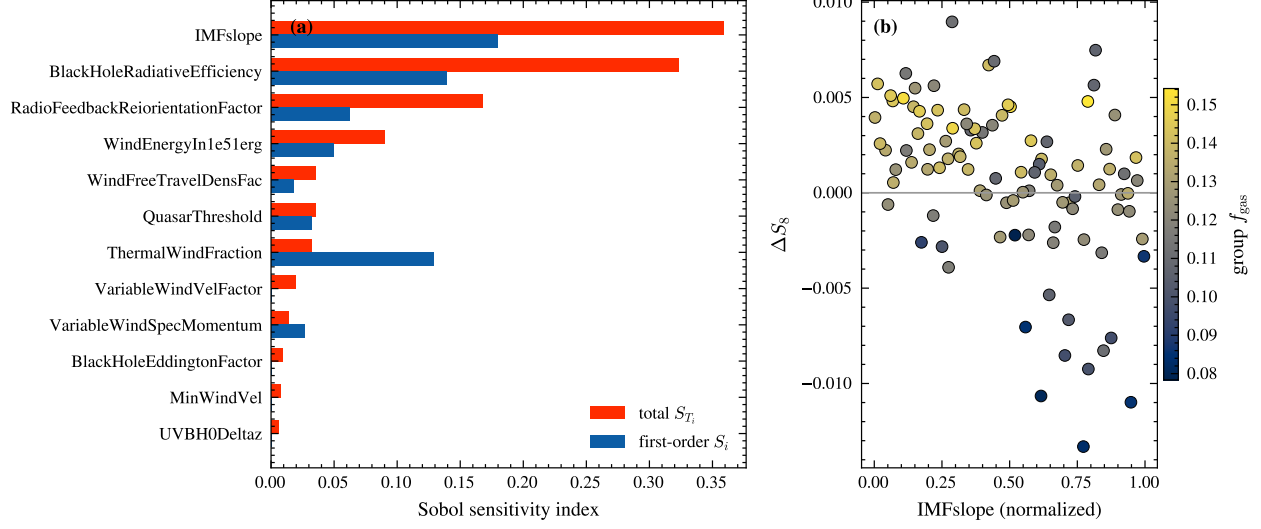


Figure 6: Left: ranked total-order (S_{T_i}) and first-order (S_i) Sobol sensitivity indices of ΔS_8 over the twelve most important of the 30 astrophysical parameters (99 runs, $\ell_{\text{max}} = 3000$). IMFslope and BlackHoleRadiativeEfficiency dominate. Right: ΔS_8 versus the normalized top driver, IMFslope, colored by group f_{gas} , showing that the top-ranked single parameter still leaves substantial scatter driven by the remaining feedback sector.

[TODO: regenerate: no cached final-product array – `lightcone_selfcal.py` computes the joint $(S_8, \Omega_m, A_{\text{bary}})$ Fisher forecast fresh at plot time via `pyccl` and never calls `np.savez`; re-running the Fisher/pyccl calculation is out of scope for this figure-integration pass (see [fig_scripts/UNREGENERATED.md](#))]

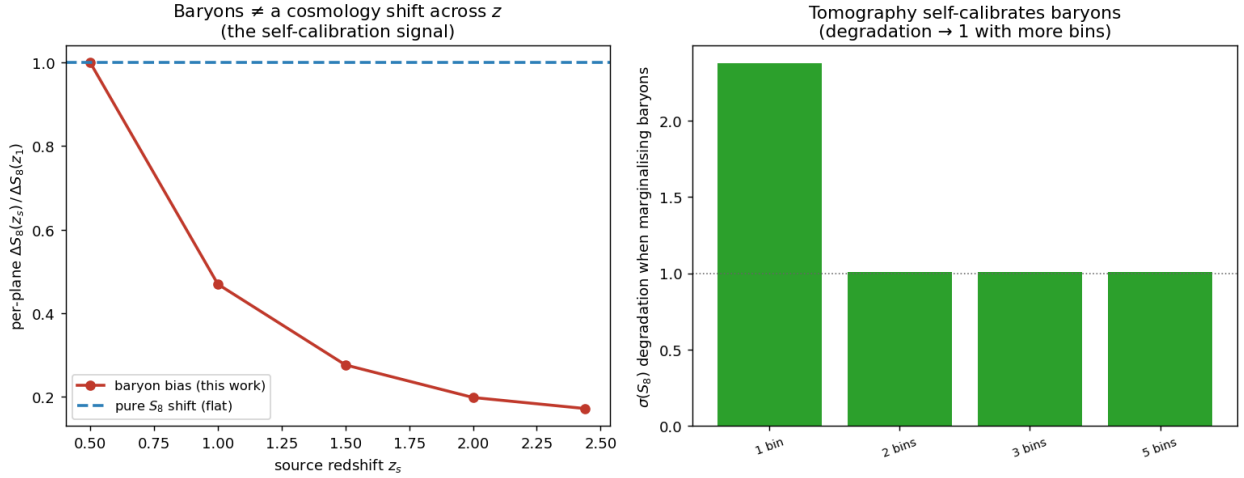


Figure 7: The corrected tomographic self-calibration result. Left: the per-plane baryonic bias $\Delta S_8(z_s)$, normalized to its value at the *first* source plane ($z_s = 0.5$; the axis label’s subscript “1” denotes the first bin index, not the redshift value $z_s = 1.0$, where the curve itself reads ≈ 0.47), declines steadily with source redshift, distinguishing it from a flat pure S_8 -shift line – the signature that lets tomography separate baryons from cosmology. Right: the $\sigma(S_8)$ degradation from marginalizing the baryon amplitude A_{bary} , which drops from $2.37\times$ with a single tomographic bin to $\approx 1\times$ with two or more well-separated bins.

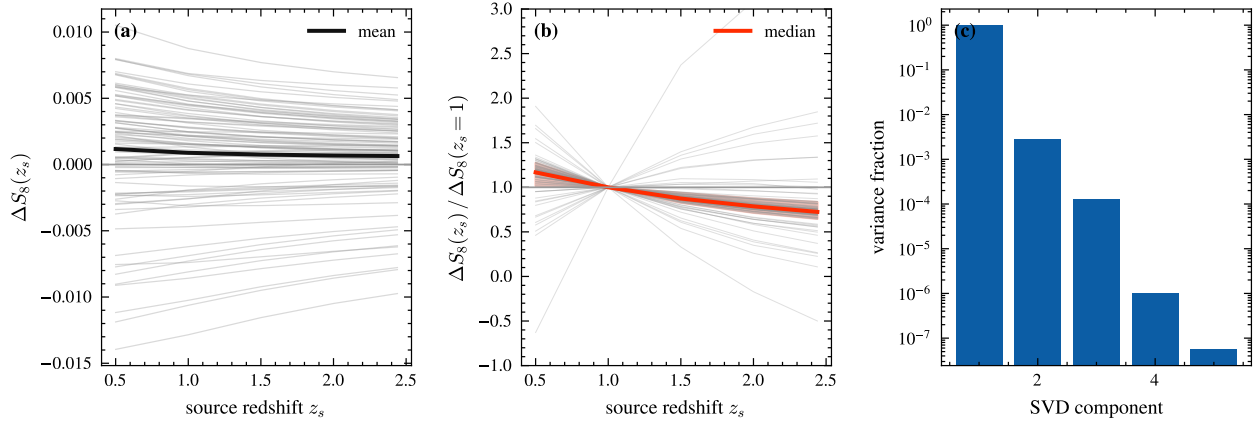


Figure 8: The original, pre-correction Step-4 self-calibration analysis, retained here only as a methodological cautionary figure. Collapsing each of the five source planes to a single amplitude (left, middle) produces a singular-value spectrum dominated by one mode, PC1 $\approx 99.7\%$ of the variance (right) – a result of the single-lightcone, nested-source-plane construction rather than a physical statement about self-calibratability. **This figure is superseded by Figure 7 and is not the paper’s self-calibration result.**

[TODO: regenerate: no cached final-product array – the BCM/van Daalen/BCemu analytic-model spans in `lightcone_model_comparison.py` are computed fresh at plot time via `pyccl` (BaryonsSchneider15, BaryonsvanDaalen19) and the external BCemu package, with no `np.savez` call; only the BIND envelope half is cached (`transfer_Sell.npz`), which is insufficient to reproduce the three-way comparison; re-running the analytic engines is out of scope for this figure-integration pass (see `fig_scripts/UNREGENERATED.md`)]

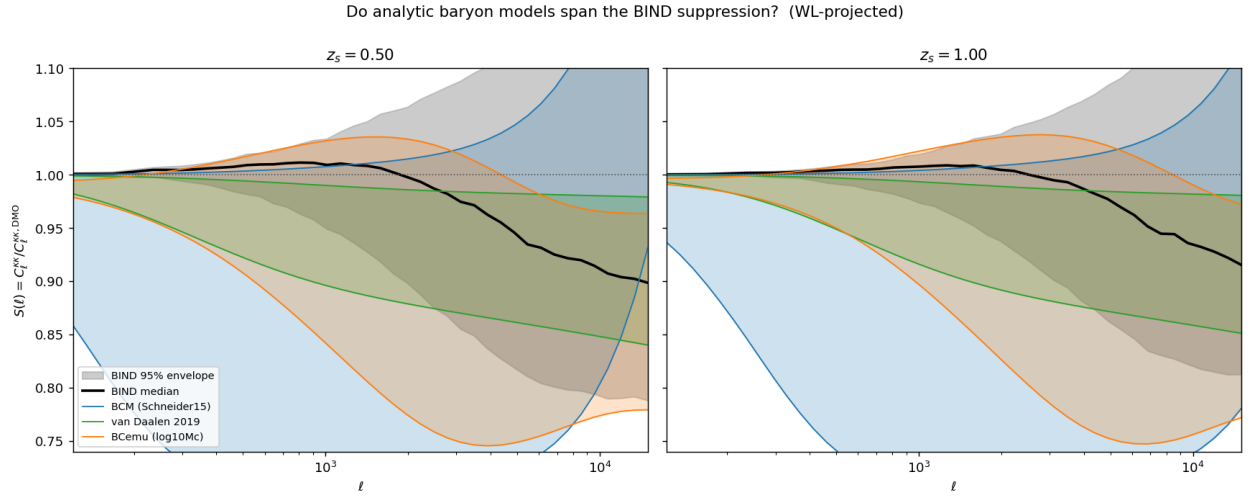


Figure 9: BIND’s median suppression and 95% envelope (black/grey) compared to the achievable span of three analytic baryon-correction models – BCM (blue, [Schneider and Teyssier, 2015](#)), van Daalen (green, [van Daalen et al., 2020](#)), and BCemu (orange) – at $z_s = 0.5$ and $z_s = 1.0$, all projected through the same weak-lensing kernel. The BIND envelope’s upper bound at $\ell \sim 10^3$, produced by gas-rich, weak-feedback enhancement, exceeds all three analytic models’ upper bounds there; at $\ell \gtrsim 5000$ the BCM (blue) span itself rises steeply and exceeds the BIND envelope, an extrapolation artifact of the Schneider15 parametrization rather than a genuine match to BIND’s enhancement corner (see Section 3.7).

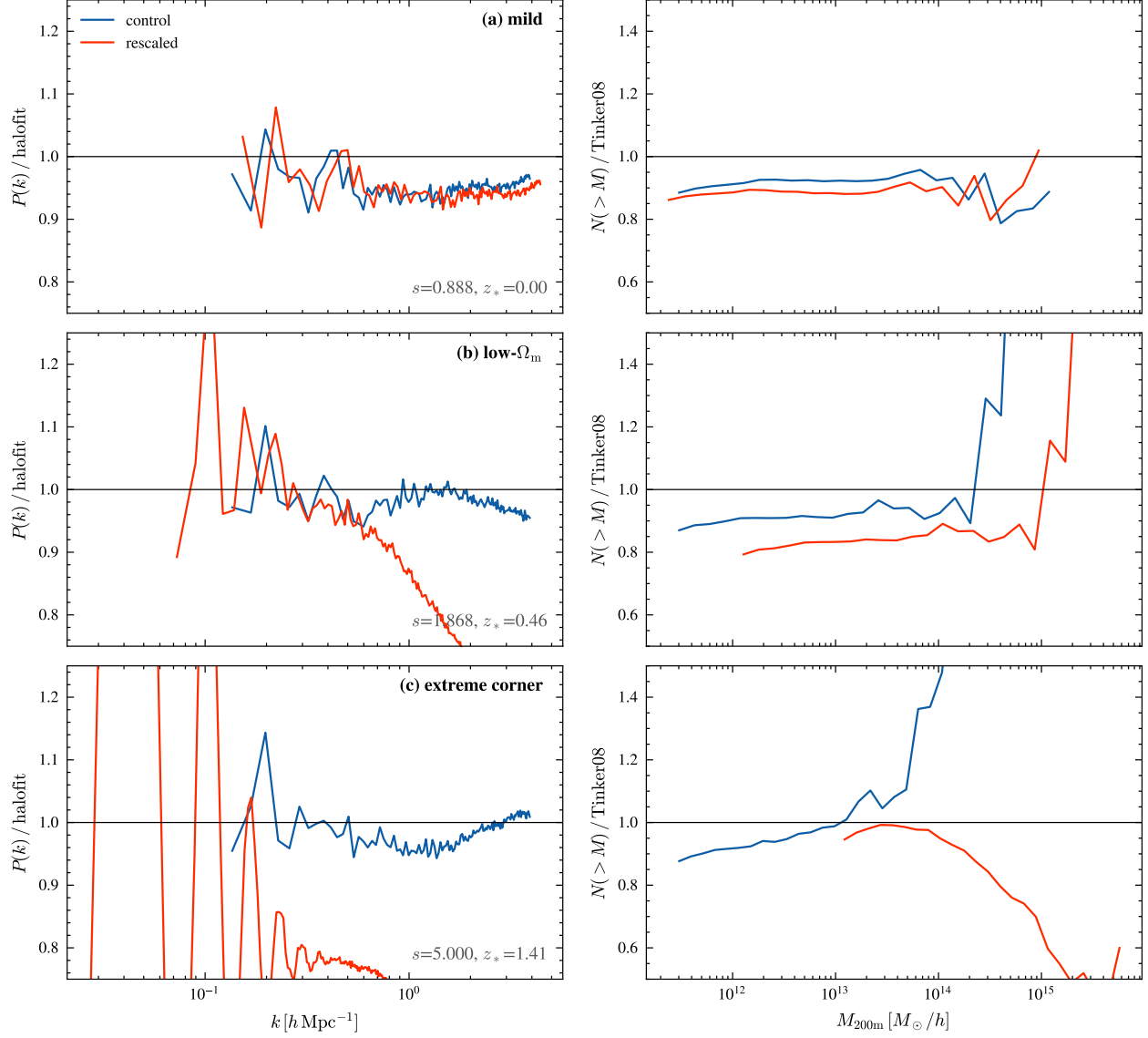


Figure 10: Real-data validation of Angulo-White rescaling: TNG300-3-Dark rescaled to three SB35 target cosmologies (rows: mild, low- Ω_m , extreme corner) versus `halofit`/`Tinker08` theory at the target (columns: $P(k)$ ratio, cumulative halo mass function ratio). Blue is the unrescaled-simulation-versus-its-own-cosmology control; red is the rescaled simulation versus the target cosmology. The mild target validates to a few percent; the extreme corner, where the solver pins its rescaling-factor bound, breaks down badly and marks the edge of the method’s validity.